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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Dsaise
Selected Domain	Domain 1: Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (primary); SDG 4.4 (skills for employment)
Project / Solution Name	HalaTuju
Name (Leader)	Leow Shen En
Email & Phone (Leader)	leowshenen@gmail.com | 018-2335269
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Mid-career workers need specific routes into durable jobs as AI changes tasks inside their current roles. A sample worker in a Penang shared-services centre has six hours a week to learn and a six-month redeployment window. She needs to know which skills still transfer, which new role is realistic, and whether the advice will remain sound as AI capabilities change. Her HR team must also compare openings, training time and cost. One-off career assessments and job-title matching miss both transferable skills and changing task exposure. This matters at scale: DOSM recorded 1,360,100 Malaysian clerical wage recipients in 2025, a broad group that includes similar workers (OpenDOSM, Annual Salaries and Wages by Occupation and Sex, 2025). The ILO refined GenAI exposure index reports especially high exposure for clerical work (Working Paper 140, 2025). These figures describe context, not a measured displacement rate. HalaTuju aims to support redeployment and relevant upskilling under SDG targets 8.5 and 4.4.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

HalaTuju turns a CV or pasted text into a living career map. Local OCR reads scans; a privacy scrubber removes personal details before Gemini or RunPod embeddings receive text. Gemini and bge-m3 match evidence in the CV against ESCO skills. On 36 held-out synthetic CVs, the current job was identified correctly in 97.2% of cases; the revised scrubber removed 99.4% of planted personal details. The map shows transferable skills, gaps, and Fastest, Balanced and Safest routes fitted to weekly study hours, with course links and RM costs. A separate full-ESCO search covers 3,039 occupations. Fixed formulas, rather than the language model, calculate matches and routes. Each plan is stress-tested in 1,000 simulated futures. A daily researcher reviews vetted sources, checks quotations, and passes claims through a second AI reviewer and fixed rules. Route-changing updates require a person; approved changes visibly reroute the map and can be undone. The prototype connects GenAI APIs, PostgreSQL database design, and cloud deployment through a Node and Express app deployed on Replit. Its sample catalogue and synthetic benchmark are labelled as such.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Browser UI: JavaScript, HTML/CSS and a 3D career map. API: Node 22, TypeScript, Express 5; local Tesseract OCR and a deterministic scoring engine. AI: Google Gemini API; BAAI/bge-m3 embeddings on RunPod Serverless; a team-trained bge-m3 plus ridge exposure estimator trained on a RunPod GPU pod. Data: PostgreSQL versioned knowledge base; ESCO, O*NET, ILO, DOSM, MyCOL and cited research. Hosting: Replit (Reserved VM) and Replit PostgreSQL 16; GitHub Actions checks. Flow: CV → local OCR and privacy scrubber → Gemini and RunPod skill matching → PostgreSQL → fixed scoring and routing → browser map. Vetted market sources → quotation check and AI review → deterministic triage → human approval for route changes → new knowledge version and reroute. (System block diagram: Appendix A.) Sponsor resources used: Yes. RunPod training and runtime embeddings; Replit hosting for the app, its PostgreSQL database and Secrets. Sign-in: Replit-managed Clerk, with guest access. The live Replit app is at https://halatuju.replit.app.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We framed the task around a sample mid-career accounts-payable worker with six hours a week and a six-month redeployment deadline. The core user questions are which skills transfer, where to go, and whether a plan survives changing AI capabilities. Early HTML prototypes led to a visible map and route choices rather than an uncheckable chat answer. We tested CV extraction and privacy against held-out synthetic cases, then improved the scrubber after the blind first result. The repo contains 418 automated tests, a type-check and documented evaluation artifacts. Human approval gates route-changing evidence, and labelled fallbacks cover AI outages. The intended Phase 2 test is a consent-based pilot with real users and verified employer opportunities; the present catalogue remains partly sample data. The app is live on Replit at https://halatuju.replit.app.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.5: full and productive employment and decent work; SDG 4.4: skills for employment. The routes also support SDG 8.2 through technological upgrading.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Context baseline: 1,360,100 Malaysian clerical wage recipients in 2025 (DOSM/OpenDOSM, wages_occupation). Product baseline on a held-out synthetic set: 97.2% correct current-job identification (35/36); 99.4% planted personal details removed after fixes. These are test results, not measured labour impact.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Track (1) % correct current-job identifications on new CVs; (2) % planted personal details removed before AI calls; (3) % users who finish a costed route within 60 seconds; (4) % pilot participants with an approved redeployment plan by their deadline.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 targets: ≥95% current-job identification and ≥99% PII removal on a fresh blind set; ≥80% of pilot users finish a route in 60 seconds; ≥70% of consenting pilot participants have an advisor-approved plan by their deadline. Pilot targets are proposed, not achieved.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js, Express, TypeScript, Tesseract, PostgreSQL libraries, 3D graph and GSAP	Code	MIT / Apache-2.0 / BSD / GreenSock terms; full inventory in technical declaration
Google Gemini and BAAI/bge-m3	Models	Gemini API terms; MIT
ESCO v1.2.1 and ESCO–O*NET crosswalk; O*NET	Datasets	European Commission reuse policy; O*NET CC BY 4.0
ILO WP140, Anthropic Economic Index, GPTs are GPTs	Research data	CC BY 4.0 / CC BY / MIT as applicable
DOSM/OpenDOSM, MyCOL and public course listings	Datasets	CC BY 4.0 / facts with attribution / linked provider facts
Clerk SDK	Code	MIT

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026

Appendix A: System block diagram

